

SQL PROJECT

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PROJECT ID: PRSQL-02-MEDICAL DATA HISTORY

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INTRODUCTION

The healthcare industry involves the generation of significant structured data on patients, physicians, hospital admissions, and medical services delivered at the regional level. The effective storage and recovery of such data are crucial for keeping track of the medical history and decision-making in the field of healthcare. SQL-based relational modeling offers a robust mechanism for managing the healthcare data in a systematic manner.

The project that catches my interest most is the Medical Data History project. The SQL application of this project is related to modeling and querying the healthcare database that comprises several interrelated tables named Patients, Doctors, Admissions, and Province_Name. The tables are associated with different entities of the medical setup and linked together through primary and foreign keys.

The purpose of this project is to prove the application of conceptual knowledge in SQL, encompassing concepts like join, filter, aggregation, and relationship mapping to a real-world healthcare setting. The database, through well-organized queries, will allow for the processing of patient admission information, doctor engagement, and provincial healthcare information, thus establishing the relevance of using SQL in healthcare data management.

SQL QUERIES

We have solved the provided list of questions as a project and attached the screenshot of the solution:

Q.1. Show first name, last name, and gender of patients whose gender is 'M'

Query:

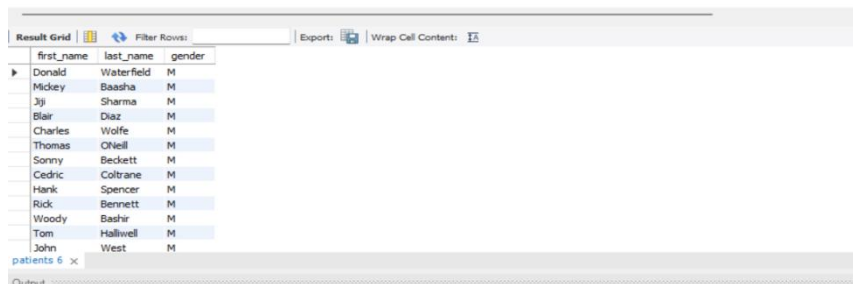
```
SELECT * FROM patients;
```

```
SELECT first_name, last_name, gender
```

```
FROM patients
```

```
WHERE gender='M'
```

```
2 • select * from patients;  
3 • select first_name , last_name, gender from patients where gender="M" ;
```



The screenshot shows a database query result grid with columns: first_name, last_name, and gender. The results are as follows:

first_name	last_name	gender
Donald	Waterfield	M
Mickey	Baasha	M
Jiji	Sharma	M
Blair	Diaz	M
Charles	Wolfe	M
Thomas	O'Neill	M
Sonny	Beckett	M
Cedric	Coltrane	M
Hank	Spencer	M
Rick	Bennett	M
Woody	Bashir	M
Tom	Hallivell	M
John	West	M

patients 6 x

Q.2. Show first name and last name of patients who does not have allergies.

Query:

```
SELECT first_name, last_name
```

```
FROM patients
```

```
WHERE allergies is null;
```

```
4 • select first_name , last_name from patients where allergies is null ;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	first_name	last_name
▶	Donald	Waterfield
	Blair	Diaz
	Thomas	O'Neill
	Sonny	Beckett
	Cedric	Coltrane
	Hank	Spencer
	Sara	di Marco
	Amy	Leela
	Rachel	Winterbourne
	John	West
	Jon	Doggett
	Angel	Edwards
	Brodie	Beck

patients 11 x

Q.3. Show first name of patients that start with the letter 'C'

Query:

SELECT *

FROM patients

WHERE first_name LIKE 'C%';

```
4 • select * from patients where first_name LIKE "C%";
```

Result Grid										Filter Rows:	Edit:	Export/Imports:	Wrap Cell Contents:
patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight				
5	Charles	Wolfe	M	2017-11-19	Orillia	ON	Penicillin	47	10				
10	Cedric	Coltrane	M	1961-11-10	Toronto	ON	Penicillin	157	61				
39	Charles	Andonuts	M	2016-05-20	Hamilton	ON	Penicillin	62	15				
45	Cross	Gordon	M	2009-03-20	Ancaster	ON	Penicillin	125	53				
63	Calleigh	Dean	F	2003-07-06	Hamilton	ON	Penicillin	175	56				
67	Catherine	Minoru	F	1997-02-25	Victoria	BC	Sulphur	184	100				
88	Caroline	Smart	F	1963-04-07	Hamilton	ON	Penicillin	145	64				
103	Casanova	Dresden	M	1987-12-20	Milton	ON	Penicillin	150	74				
117	Chen	Baasha	M	1992-05-09	Cambridge	ON	Penicillin	193	140				
152	Charman	Hardy	F	1992-04-16	Hamilton	ON	Penicillin	155	93				
188	Cary	Henderson	M	1961-07-31	Hamilton	ON	Bee St...	167	98				
192	Christine	Lynch	F	1994-08-01	Brantford	ON	Sulphur	181	77				
211	Carol	Wyle	F	1981-08-04	Toronto	ON	Penicillin	170	100				

patients 17

```
4 • select FIRST_NAME FROM patients WHERE FIRST_NAME LIKE "C%";
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
FIRST_NAME			
Charles			
Cedric			
Charles			
Cross			
Calleigh			
Catherine			
Caroline			
Casanova			
Chen			
Charmian			
Cary			
Christine			
Carol			

patients 18 x

Q.4. Show first name and last name of patients that weight within the range of 100 to 120 (inclusive).

Query:

```
SELECT first_name, last_name, weight
```

```
FROM patients
```

```
WHERE weight BETWEEN 100 AND 120;
```

```
4 • select FIRST_NAME , last_name , weight FROM patients WHERE weight between 100 and 120 ;
```

FIRST_NAME	last_name	weight
Ethan	Maureen	120
Gary	Graf	120
Alan	Hardbroom	120
Avon	Tony	120
Arnold	Tate	120
Tom	Lopez	120
Artemus	Freeman	120
Nino	Haruko	120
Ronnie	Sanderson	120
Domenico	Glicker	120
Jill	Jorkins	120
William	Scotts	120
Joe	Disher	120

patients 21 x

Q.5. Update the patients table for the allergy's column. If the patient's allergies is null then replace it with 'NKA'

Query:

SELECT *

CASE WHEN allergies IS NULL THEN 'NKA'

ELSE allergies

END AS allergies

FROM patients;

```
4 • select * ,
5   case when allergies is null then "NKA"
6   else allergies
7   end as allergies
8   from patients ;
```

patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight	allergies
1	Donald	Waterfield	M	1963-02-12	Barrie	ON	XXXX	156	65	NKA
2	Mickey	Bassha	M	1981-05-28	Dundas	ON	Sulfa	185	76	Sulfa
3	Jji	Sharma	M	1957-09-05	Hamilton	ON	Penicillin	194	106	Penicillin
4	Eliar	Diaz	M	1967-01-07	Hamilton	ON	XXXX	191	104	NKA
5	Charles	Wolfe	M	2017-11-19	Orillia	ON	Penicillin	47	10	Penicillin
6	Sue	Falcon	F	2017-09-30	Ajax	ON	Penicillin	43	5	Penicillin
7	Thomas	O'Neill	M	1993-01-31	Burlington	ON	XXXX	180	117	NKA
8	Sonny	Beckett	M	1952-12-11	Port Hawkesbury	NS	XXXX	174	105	NKA
9	Sister	Spitzer	F	1966-10-15	Toronto	ON	Penicillin	173	95	Penicillin
10	Cedric	Coltrane	M	1961-11-10	Toronto	ON	XXXX	157	61	NKA
11	Hank	Spencer	M	1969-08-10	Peterborough	ON	XXXX	158	74	NKA
12	Sara	d'Marco	F	1949-04-29	Hamilton	ON	XXXX	145	46	NKA
13	Daphne	Seabright	F	1954-11-18	Ancaster	ON	Codene	146	77	Codene

Result 24 x

Q.6. Show first name and last name concatenated into one column to show their full name.

Query:

```
SELECT *
```

```
FROM admissions;
```

```
SELECT concat (first_name,' ', last_name) AS full_name
```

```
FROM patients;
```

```
3 • select * from admissions ;  
4 • select concat(first_name,' ',last_name) as full_name from patients ;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	full_name			
▶	Donald Waterfield			
	Mickey Baasha			
	Jiji Sharma			
	Blair Diaz			
	Charles Wolfe			
	Sue Falcon			
	Thomas O'Neill			
	Sonny Beckett			
	Sister Spitzer			
	Cedric Coltrane			
	Hank Spencer			
	Sara di Marco			
	Daphne Seabright			

Result 30 x

Q.7. Show first name, last name, and the full province name of each patient.

Query:

```
SELECT * FROM province_name;
```

```
SELECT p.first_name, p.last_name, p1.province_name
```

```
FROM patients;
```



```

4 • select * from province_names ;
5 • select p.first_name,p.last_name , p1.province_name |from patients p
6   left join province_names p1
7   on p.province_id=p1.province_id ;

```

first_name	last_name	province_name
Donald	Waterfield	Ontario
Mickey	Baasha	Ontario
Jji	Sharma	Ontario
Blair	Diaz	Ontario
Charles	Wolfe	Ontario
Sue	Falcon	Ontario
Thomas	O'Neill	Ontario
Sonny	Beckett	Nova Scotia
Sister	Spitzer	Ontario
Cedric	Coltrane	Ontario
Hank	Spencer	Ontario
Sara	di Marco	Ontario
Daphne	Seabright	Ontario

Q.8. Show how many patients have a birth_date with 2010 as the birth year.

Query_1:

SELECT COUNT (*) AS birth_date

FROM patients

WHERE year(birth_date) =2010;

```

6 • select count(*) as birth_date from patients where year(birth_date)=2010;

```

birth_date
55

Query_2:

SELECT COUNT (*) AS birth_date

FROM patients

WHERE birth_date LIKE "2010-%-%";

```
6 • select count(*) as birth_date from patients where birth_date like "2010-%-";
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
birth_date			
55			

Q.9. Show the first_name, last_name, and height of the patient with the greatest height.

Query:

```
SELECT first_name, last_name, height
```

```
FROM patients
```

```
ORDER BY height DESC;
```

```
6 • select first_name, last_name, height from patients order by height desc ;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
first_name	last_name	height	
Sam	Haruko	226	
Peter	Drake	225	
Joe	Snyder	224	
Victor	Fleming	224	
Joe	Brady	224	
James	Reilly	223	
James	Dundee	223	
Martin	Overstreet	223	
Jake	Stanfield	223	
Jerry	Camonte	223	
Alberto	Bradley	222	
Thomas	Smart	222	
Primo	Harding	222	

patients 52 x

Q.10. Show all columns for patients who have one of the following patient_ids:

1,45,534,879,1000

Query:

```
SELECT *
```

FROM patients

WHERE patient_id IN (1, 45, 534, 879, 1000);

```
6 • select * from patients where patient_id in(1 ,45 ,534 ,879 ,1000) ;
```

	patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight
▶	1	Donald	Waterfield	M	1963-02-12	Barrie	ON	HULL	156	65
	45	Cross	Gordon	M	2009-03-20	Ancaster	ON	HULL	125	53
	534	Don	Zatara	M	2008-01-11	Timmins	ON	HULL	136	67
	879	Orla	Shawn	F	1967-09-24	Sarnia	ON	Penicillin	149	65
	1000	Rick	Williams	M	1975-04-13	Hamilton	ON	Penicillin	176	127
*	HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL

Q.11. Show the total number of admissions.

Query:

SELECT COUNT (*) AS total_admissions

FROM admissions;

```
6 • select count(*) as total_admissions from admissions ;
```

	total_admissions
▶	5067

Q.12. Show all the columns from admissions where the patient was admitted and discharged on the same day.

Query:

```
SELECT *
```

```
FROM admissions
```

```
WHERE admission_date= discharge_date;
```

```
6 • select * from admissions where admission_date=discharge_date ;
7
```

patient_id	admission_date	discharge_date	diagnosis	attending_doctor_id
1	2018-09-20	2018-09-20	Ineffective Breathing Pattern R/T Fluid Accumulation	24
9	2018-12-31	2018-12-31	Ruptured Appendicitis	19
10	2019-02-27	2019-02-27	Lower Quadrant Pain	27
17	2019-03-04	2019-03-04	Diabetes Mellitus	9
28	2019-03-30	2019-03-30	Cancer Of The Stomach	26
31	2018-09-26	2018-09-26	Cardiovascular Disease	19
53	2018-10-24	2018-10-24	Urinary Tract Infection	8
54	2019-04-07	2019-04-07	Hypertension	21
70	2018-07-17	2018-07-17	Migraine	20
78	2018-06-17	2018-06-17	Hypertension	17
91	2018-08-30	2018-08-30	Congestive Heart Failure	3
92	2019-01-03	2019-01-03	Osteoarthritis Knee	5
93	2019-05-20	2019-05-20	Left Cerebral Vascular Accident	12

Q.13. Show the total number of admissions for patient_id 579.

Query:

```
SELECT COUNT (*) AS total_numbers
```

```
FROM admissions
```

```
WHERE patient_id= 579;
```

```
6 • select count(*) as total_numbers from admissions where patient_id = 579;
```

total_numbers
2

Q.14. Based on the cities that our patients live in, show unique cities that are in province_id 'NS'?

Query:

```
SELECT DISTINCT (p.city), p1.province_name, p1.province_id
```

```
FROM patients p;
```

```
6 • select distinct(p.city) , p1.province_name , p1.province_id from patients p
7   join province_names p1
8   on p.province_id =p1.province_id
9   where p1.province_id ="NS" ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	city	province_name	province_id
▶	Port Hawkesbury	Nova Scotia	NS
	Halifax	Nova Scotia	NS
	Inverness	Nova Scotia	NS

Q.15. Write a query to find the first_name, last name and birth date of patients who have height more than 160 and weight more than 70?

Query:

```
SELECT first_name, last_name, birth_date
```

```
FROM patients
```

```
WHERE height>=160 AND weight>=70;
```

```
6 • select first_name , last_name , birth_date from patients where height >=160 and weight >=70 ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	first_name	last_name	birth_date
▶	Mickey	Baasha	1981-05-28
	Jiji	Sharma	1957-09-05
	Blair	Diaz	1967-01-07
	Thomas	O'Neill	1993-01-31
	Sonny	Beckett	1952-12-11
	Sister	Spitzer	1966-10-15
	Rick	Bennett	1977-01-27
	Amy	Leela	1977-06-25
	Tom	Halliwel	1987-08-01
	Rachel	Winterbourne	1966-04-26
	Jon	Doggett	1951-12-25
	Angel	Edwards	1975-08-22
	Nino	Andrews	2001-04-21

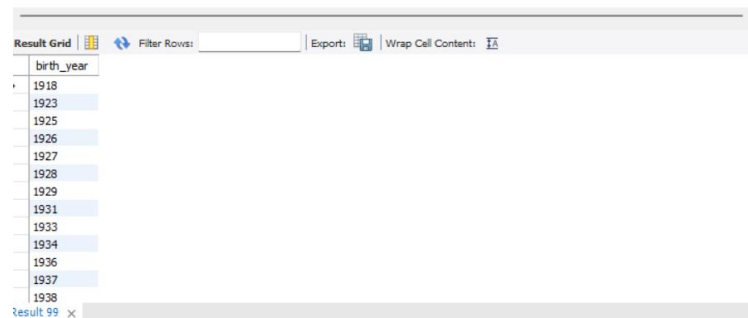
patients 92 x

Q.16. Show unique birth years from patients and order them by ascending.

Query:

```
SELECT DISTINCT year(birth_date) AS birth_year  
FROM patients  
ORDER BY birth_year ASC;
```

```
6 • select distinct year(birth_date) as birth_year from patients order by birth_year asc;
```



The screenshot shows a database query result grid. The header row is labeled 'birth_year'. The data rows list years from 1918 to 1938, with some years missing (1919, 1921, 1924, 1930, 1932, 1935, 1939). The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Contents' option.

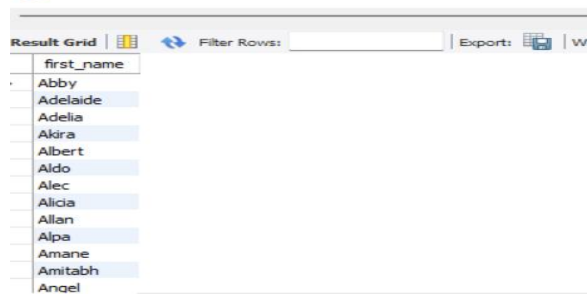
birth_year
1918
1923
1925
1926
1927
1928
1929
1931
1933
1934
1936
1937
1938

Q.17. Show unique first names from the patients table which only occurs once in the list.

Query:

```
SELECT first_name  
FROM patients  
GROUP BY first_name  
HAVING COUNT (*) = 1;
```

```
6 • SELECT first_name  
7 FROM patients  
8 GROUP BY first_name  
9 HAVING COUNT (*) = 1;  
10
```



The screenshot shows a database query result grid. The header row is labeled 'first_name'. The data rows list 15 unique first names: Abby, Adelaide, Adelia, Akira, Albert, Aldo, Alec, Alicia, Allan, Alpa, Amane, Amitabh, and Angel. The interface includes a 'Filter Rows' field, an 'Export' button, and a 'Wrap Cell Contents' option.

first_name
Abby
Adelaide
Adelia
Akira
Albert
Aldo
Alec
Alicia
Allan
Alpa
Amane
Amitabh
Angel

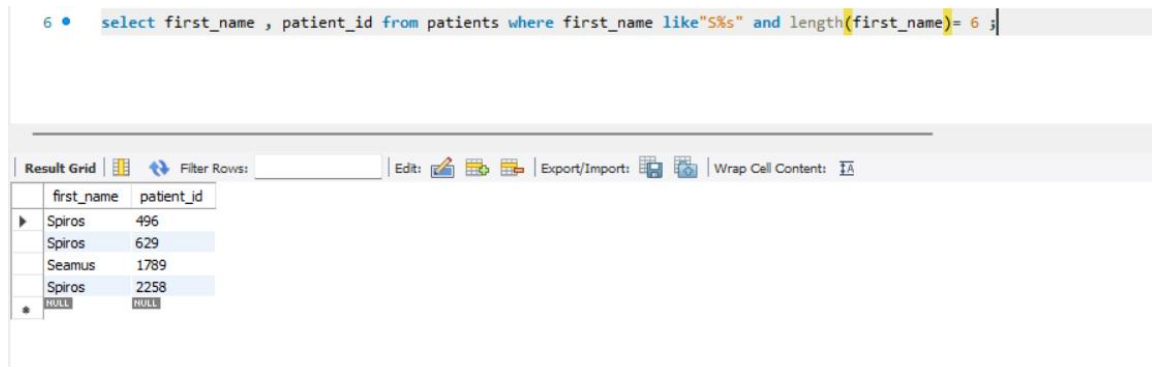
Q.18. Show patient_id and first_name from patients where their first_name start and ends with 's' and is at least 6 characters long.

Query:

SELECT first_name, patient_name,

FROM patients

WHERE first_name LIKE 'S%s' AND length(first_name) = 6;



```
6 • select first_name , patient_id from patients where first_name like "S%s" and length(first_name) = 6 ;
```

first_name	patient_id
Spiros	496
Spiros	629
Seamus	1789
Spiros	2258
MURPHY	MURPHY

Q.19. Show patient_id, first_name, last_name from patients whose diagnosis is 'Dementia'. Primary diagnosis is stored in the admissions table.

Query:

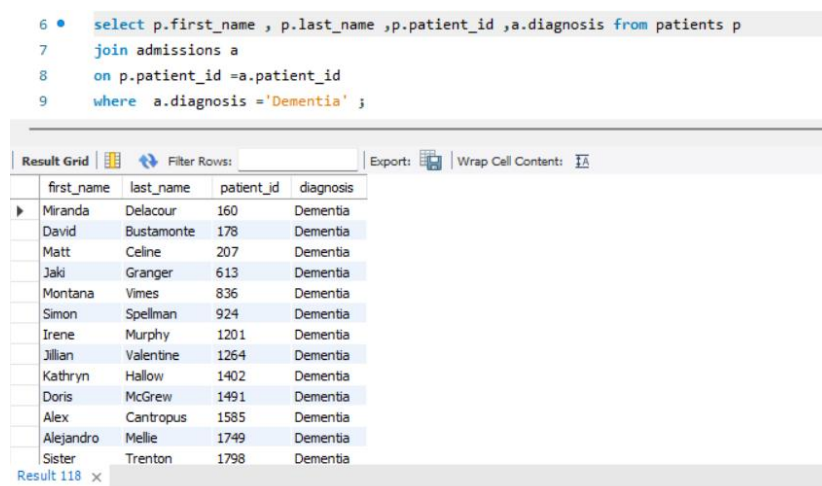
SELECT p.first_name, p.last_name, p.patient_id, a.diagnosis

FROM patients p

JOIN admissions a

ON p.patient_id = a.patient_id

WHERE a.diagnosis = 'Dementia';



```
6 • select p.first_name , p.last_name , p.patient_id , a.diagnosis from patients p
7 join admissions a
8 on p.patient_id = a.patient_id
9 where a.diagnosis = 'Dementia' ;
```

first_name	last_name	patient_id	diagnosis
Miranda	Delacour	160	Dementia
David	Bustamonte	178	Dementia
Matt	Celine	207	Dementia
Jaki	Granger	613	Dementia
Montana	Vimes	836	Dementia
Simon	Spellman	924	Dementia
Irene	Murphy	1201	Dementia
Jillian	Valentine	1264	Dementia
Kathryn	Hallow	1402	Dementia
Doris	McGrew	1491	Dementia
Alex	Cantropus	1585	Dementia
Alejandro	Mellie	1749	Dementia
Sister	Trenton	1798	Dementia

Q.20. Display every patient's first_name. Order the list by the length of each name and then by alphabetically.

Query:

SELECT first_name

FROM patients

ORDER BY length(first_name) ASC, first_name ASC;

6 • `select first_name from patients order by length(first_name) asc , first_name asc ;`

first_name
Al
Al
Al
Al
Al
Al
Al
Bo
Bo
Bo
Bo
Bo
Bo

Q.21. Show the total amount of male patients and the total amount of female patients in the patient table. Display the two results in the same row.

Query:

SELECT gender, COUNT (*)

FROM patients

GROUP BY gender;

6 • `SELECT GENDER , COUNT(*) FROM patients GROUP BY GENDER ;`

GENDER	COUNT(*)
F	2062
M	2468

Q.22. Show the total amount of male patients and the total amount of female patients in the patient table. Display the two results in the same row.

Query:

```
SELECT gender, COUNT (*)
```

```
FROM patients
```

```
GROUP BY gender;
```

```
6 • SELECT GENDER , COUNT(*) FROM patients GROUP BY GENDER ;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	GENDER	COUNT(*)		
▶	F	2062		
	M	2468		

Q.23. Show patient_id, diagnosis from admissions. Find patients admitted multiple times for the same diagnosis.

Query:

```
SELECT p.patient_id, a.diagnosis
```

```
FROM patients p
```

```
JOIN admissions a
```

```
ON p.patients_id = a.patinet_id
```

```
GROUP BY patient_id, diagnosis
```

```
HAVING COUNT (*)>1;
```

```

6 • select p.patient_id ,a.diagnosis from patients p
7   join admissions a
8   on p.patient_id = a.patient_id
9   GROUP BY patient_id, diagnosis
10  HAVING COUNT(*) > 1;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
patient_id	diagnosis			
137	Pregnancy			
320	Pneumonia			
1577	Congestive Heart Failure			
2004	Left Shoulder Rotator Cuff Repair			
2859	Severed Spine At C3			
3012	Appendicitis			
3367	Pyelonephritis			
3468	Congestive Heart Failure			
4083	Congestive Heart Failure			
4121	Congestive Heart Failure			
4363	Congestive Heart Failure			

Q.24. Show the city and the total number of patients in the city. Order from most to least patients and then by city name ascending.

Query:

SELECT city, COUNT (*)

FROM patient

GROUP BY city

ORDER BY COUNT (*) DESC, city ASC;

```

6 • select city , count(*) from patients group by city order by count(*) desc , city asc ;
7

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
city	count(*)			
Hamilton	1938			
Toronto	317			
Burlington	276			
Brantford	147			
Ancaster	117			
Stoney Creek	107			
Cambridge	79			
Dundas	79			
Milton	65			
Paris	58			
...	...			

Q.25. Show first name, last name and role of every person that is either patient or doctor.
The roles are either "Patient" or "Doctor".

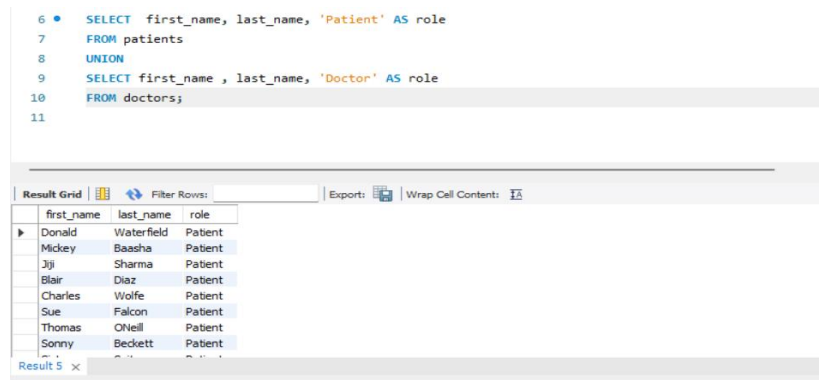
```
SELECT first_name, last_name, 'Patient' AS role
```

```
FROM patient
```

```
UNION
```

```
SELECT first_name, last_name, 'Doctor' AS role
```

```
FROM doctors;
```



The screenshot shows a SQL query editor with a query that uses a UNION to combine data from two tables. The query is as follows:

```
6 • SELECT first_name, last_name, 'Patient' AS role
7 FROM patients
8 UNION
9 SELECT first_name, last_name, 'Doctor' AS role
10 FROM doctors;
11
```

Below the query editor, a "Result Grid" is displayed, showing the results of the query. The grid has three columns: first_name, last_name, and role. The results are as follows:

first_name	last_name	role
Donald	Waterfield	Patient
Mickey	Baasha	Patient
Jiji	Sharma	Patient
Blair	Diaz	Patient
Charles	Wolfe	Patient
Sue	Falcon	Patient
Thomas	O'Neill	Patient
Sonny	Beckett	Patient

Q.26. Show all allergies ordered by popularity. Remove NULL values from query.

Query:

```
SELECT COUNT (*), diagnosis
```

```
FROM admissions
```

```
WHERE diagnosis IS NOT NULL
```

```
GROUP BY diagnosis
```

```
ORDER BY COUNT (*) DESC;
```

```

6 • select count(*) , diagnosis from admissions where diagnosis is not null
7 |group by diagnosis order by count(*) desc ;

```

count(*)	diagnosis
277	Congestive Heart Failure
170	Pregnancy
142	Appendicitis
105	Myocardial Infarction
83	Chest Pain
82	Pneumonia
68	Unstable Angina
61	Fractured Femur
58	Abdominal Pain
50	Asthma
44	...

Q.27. Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.

Query:

```
SELECT first_name, last_name, birth_date
```

```
FROM patients
```

```
WHERE birth_date BETWEEN '1970-01-01' AND '1979-12-31'
```

```
ORDER BY birth_date ASC;
```

```

6 • SELECT first_name,last_name,birth_date
7 FROM patients
8 WHERE birth_date BETWEEN '1970-01-01' AND '1979-12-31'
9 ORDER BY birth_date ASC;
10

```

first_name	last_name	birth_date
Frances	Kobayakawa	1970-01-02
Sunny	Burrell	1970-01-07
Penelope	Beckett	1970-01-14
Deborah	Stewart	1970-01-14
Augusta	Decker	1970-01-22
Sookie	Brearily	1970-02-01
Temple	Wylie	1970-02-10
Deanna	Spano	1970-03-23
Jadu	Principal	1970-03-28
Betty	Stephens	1970-03-28
Jo	Sahid	1970-03-31
...	...	...

Q.28. We want to display each patient's full name in a single column. Their last_name in all upper letters must appear first, then first_name in all lowercase letters. Separate the last_name and first_name with a comma. Order the list by the first_name in descending order EX: SMITH, jane

Query:

```
SELECT concat (upper (last_name), ',', lower (first_name) AS full_name
```

FROM patients

ORDER BY first_name DESC;

```
6 • select concat(upper(last_name), ' ', lower(first_name)) as full_name from patients order by first_name desc ;
```



full_name
MILLER, zoe
CORBIE, jiva
KOBAYAKAWA, zenigata
OVERSTREET, zenigata
BENNETT, zen
MEPHESTO, zelda
MORRIS, zelda
THOMAS, zeffram
FLUTE, zeffram
MARONEY, zatarina

Q.29. Show the province_id (s), sum of height; where the total sum of its patient's height is greater than or equal to 7,000.

Query:

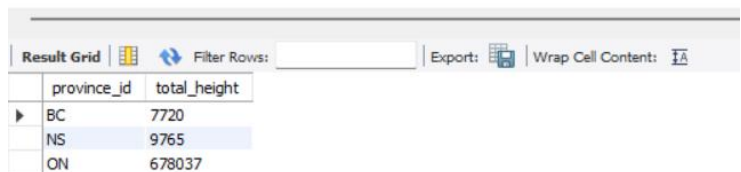
SELECT province_id, SUM (height) AS total_height

FROM patients

GROUP BY province_id

HAVING SUM (height) >= 7000;

```
6 • SELECT province_id, SUM(height) AS total_height
7 FROM patients
8 GROUP BY province_id
9 HAVING SUM(height) >= 7000;
10
```



province_id	total_height
BC	7720
NS	9765
ON	678037

Q.30. Show the difference between the largest weight and smallest weight for patients with the last name 'Maroni'.

Query:

SELECT MAX (weight) - MIN (weight) AS difference

FROM patients

WHERE last_name='Maroni';

```
6 • select max(weight) - min(weight) as difference from patients where last_name="Maroni" ;
```

difference
71

Q.31. Show all of the days of the month (1-31) and how many admission_dates occurred on that day. Sort by the day with most admissions to least admissions.

Query:

```
SELECT DAY (admission_date) AS day_of_month, COUNT (*) AS total_admissions
FROM admissions
GROUP BY DAY (admission_date)
ORDER BY total_admissions DESC;
```

```
6 • SELECT DAY(admission_date) AS day_of_month,COUNT(*) AS total_admissions
7 FROM admissions
8 GROUP BY DAY(admission_date)
9 ORDER BY total_admissions DESC;
```

day_of_month	total_admissions
11	184
4	184
9	183
2	180
6	179
12	179
16	177
21	174
13	173
28	173
7	167
5	166

Q.32. Show all of the patients grouped into weight groups. Show the total amount of patients in each weight group. Order the list by the weight group descending. e.g. if they weight 100 to109 they are placed in the 100-weight group, 110-119 = 110 weight group, etc.

Query:

```
SELECT
```

CASE

WHEN weight BETWEEN 0 AND 9 THEN 0

WHEN weight BETWEEN 10 AND 19 THEN 10

WHEN weight BETWEEN 20 AND 29 THEN 20

WHEN weight BETWEEN 30 AND 39 THEN 30

WHEN weight BETWEEN 40 AND 49 THEN 40

WHEN weight BETWEEN 50 AND 59 THEN 50

WHEN weight BETWEEN 60 AND 69 THEN 60

WHEN weight BETWEEN 70 AND 79 THEN 70

WHEN weight BETWEEN 80 AND 89 THEN 80

WHEN weight BETWEEN 90 AND 99 THEN 90

WHEN weight BETWEEN 100 AND 109 THEN 100

WHEN weight BETWEEN 110 AND 119 THEN 110

WHEN weight BETWEEN 120 AND 129 THEN 120

WHEN weight BETWEEN 130 AND 139 THEN 130

WHEN weight BETWEEN 140 AND 149 THEN 140

WHEN weight BETWEEN 150 AND 159 THEN 150

END AS weight_group,

COUNT (DISTINCT patient_id) AS total_patients

FROM patients

GROUP BY weight_group

ORDER BY weight_group DESC;

```

6 SELECT
7 CASE
8   WHEN weight BETWEEN 0 AND 9 THEN 0
9   WHEN weight BETWEEN 10 AND 19 THEN 10
10  WHEN weight BETWEEN 20 AND 29 THEN 20
11  WHEN weight BETWEEN 30 AND 39 THEN 30
12  WHEN weight BETWEEN 40 AND 49 THEN 40
13  WHEN weight BETWEEN 50 AND 59 THEN 50
14  WHEN weight BETWEEN 60 AND 69 THEN 60
15  WHEN weight BETWEEN 70 AND 79 THEN 70
16  WHEN weight BETWEEN 80 AND 89 THEN 80
17  WHEN weight BETWEEN 90 AND 99 THEN 90
18  WHEN weight BETWEEN 100 AND 109 THEN 100
19  WHEN weight BETWEEN 110 AND 119 THEN 110
20  WHEN weight BETWEEN 120 AND 129 THEN 120
21  WHEN weight BETWEEN 130 AND 139 THEN 130
22  WHEN weight BETWEEN 140 AND 149 THEN 140
23  WHEN weight BETWEEN 150 AND 159 THEN 150
24 END AS weight_group,
25 COUNT(DISTINCT patient_id) AS total_patients
26 FROM patients
27 GROUP BY weight_group
28 ORDER BY weight_group DESC;
29

```

weight_group	total_patients
140	6
130	59
120	191
110	426
100	507
90	403
80	478
70	633
60	685
50	707
40	729
30	751
20	773
10	795
0	817

Q.33. Show patient_id, weight, height, is Obese from the patient table. Display is Obese as a Boolean 0 or 1. Obese is defined as $\text{weight(kg)} / (\text{height(m)})^2 \geq 30$. Weight is in units kg. Height is in units cm.

Query:

SELECT patient_id, weight, height,

CASE

WHEN weight / power (height/100.0, 2)>=30 THEN 1

ELSE 0

END AS obese

FROM patients;

```

6 select patient_id,weight,height ,
7 case
8   when weight / power(height/100.0 ,2)>=30 then 1
9   else 0
10  end as obese
11 from patients ;

```

patient_id	weight	height	obese
1	65	156	0
2	76	185	0
3	106	194	0
4	104	191	0
5	10	47	1
6	5	43	0
7	117	180	1
8	105	174	1

Q.34. Show patient_id, first_name, last_name, and attending doctor's specialty. Show only the patients who has a diagnosis as 'Epilepsy' and the doctor's first name is 'Lisa'. Check patients, admissions, and doctors tables for required information.

Query:

```
SELECT p.patient_id, p.first_name, p.last_name, d.specialty
```

```
FROM patients p
```

```
JOIN admissions a
```

```
ON p.patient_id = a.patient_id
```

```
JOIN doctors d
```

```
ON a.attending_doctor_id = d.doctor_id
```

```
WHERE diagnosis = 'Epilepsy' AND d.first_name='Lisa';
```

```
6 • select p.patient_id ,p.first_name , p.last_name , d.specialty from patients p
7   join admissions a
8   on p.patient_id =a.patient_id
9   join doctors d
10  on a.attending_doctor_id =d.doctor_id
11  where diagnosis ="Epilepsy" and d.first_name="Lisa" ;
```

patient_id	first_name	last_name	specialty
468	Frank	Anderson	Obstetrician/Gynecologist
701	Precious	Ashton	Obstetrician/Gynecologist

INSIGHTS

1. The database contains **4,530 unique patients** and **27 doctors**, with a total of **5,067 admission records**, indicating repeat hospital visits for several patients.
2. Gender analysis shows **2,468 male patients** and **2,062 female patients**, reflecting a slightly higher male patient population.
3. The most frequently reported diagnosis is **Congestive Heart Failure (277 cases)**, followed by **Pregnancy (170 cases)** and **Appendicitis (142 cases)**.
4. A total of **481 admissions** show patients being admitted and discharged on the **same day**, indicating short-duration or emergency treatments.
5. There are **11 cases** where patients were admitted **multiple times for the same diagnosis**, suggesting chronic or recurring medical conditions.
6. City-wise analysis shows **Hamilton** has the highest number of patients (**1,938 patients**), followed by **Toronto (317)** and **Burlington (276)**.
7. Obesity analysis based on BMI calculation reveals **2,222 patients** fall under the **obese category**, highlighting a major health risk trend.
8. Age distribution analysis shows **55 patients** were born in the year **2010**, representing younger age-group admissions.
9. Province-based mapping confirms patient data is spread across **13 provinces**, enabling region-wise healthcare analysis.
10. Doctor-admission linkage shows that patient diagnoses are distributed across multiple medical specialties, supporting comprehensive treatment coverage.

CHALLENGES

Challenges faced during this project :

1. **Challenge:** Repeating Query Errors

How We Overcame: We carefully analysed SQL error messages, executed queries step-by-step, and tested individual clauses separately to identify and fix syntax and logical mistakes.

2. **Challenge:** Query Optimization and Method Selection

How We Overcame: Different approaches were evaluated, and the most efficient method was chosen by optimizing joins, reducing unnecessary subqueries, and improving overall query performance.

3. **Challenge:** Database Design and Relationship Issues

How We Overcame: Tables were redesigned using normalization techniques, and correct primary and foreign key relationships were implemented to ensure data integrity.

4. **Challenge:** Handling Inconsistent and Missing Data

How We Overcame: Data cleaning techniques such as filtering NULL values, removing duplicates, and validating data formats were applied to maintain accuracy.

5. **Challenge:** Software and Tool-Related Issues

How We Overcame: Software issues were resolved by updating the SQL environment, restarting connections, and maintaining backups to prevent data loss.

6. **Challenge:** Complex Join Conditions

How We Overcame: Join logic was tested incrementally, and table relationships were thoroughly analysed to avoid duplicate or missing records.

7. **Challenge:** Debugging Logical Errors

How We Overcame: Query results were verified using sample data and manual checks to ensure the logic produced correct and meaningful outputs.

8. **Challenge:** Learning Advanced SQL Concepts

How We Overcame: Advanced SQL features were understood through documentation, practice queries, and gradual implementation within the project.

SUMMARY

The Medical History SQL project demonstrates effective use of relational databases to store, manage, and analyse healthcare data. By integrating patient, doctor, admission, and province information, the project provides meaningful insights into patient demographics, disease patterns, hospital admissions, and regional healthcare distribution.

The analysis highlights key trends such as frequently diagnosed conditions, repeat admissions, obesity prevalence, and city-wise patient concentration. Overall, the project showcases practical application of SQL concepts including joins, aggregations, filtering, and data normalization, emphasizing the importance of structured data analysis in healthcare decision-making.